

Bayesian Optimization in a Billion Dimensions **via Random Embeddings**

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February 25, 2025

Intelligent Systems, MIPT

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Motivation

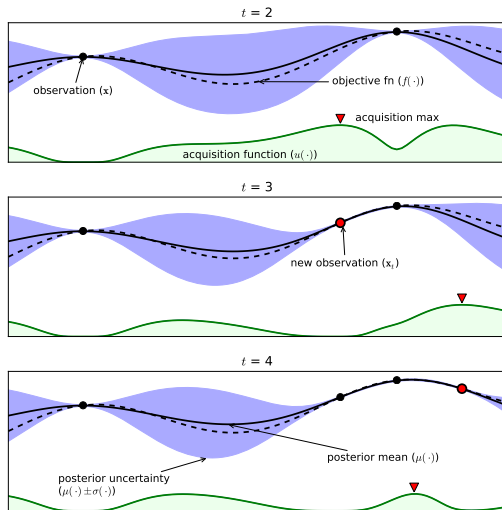
What is Bayesian Optimization?

Bayesian Optimization (BO) is a **sequential** global optimization strategy for expensive **black-box** functions $f : \mathcal{X} \rightarrow \mathbb{R}$, where $\mathcal{X} \subset \mathbb{R}^D$, and:

- f has 1) no closed form, 2) noisy evaluations, or 3) costly gradients.
- Goal: Find $\mathbf{x}^* = \arg \max_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x})$ with few function evaluations.

Key Steps of Bayesian Optimization

1. Use the prior to decide at which input $\mathbf{x} \in \mathcal{X}$ to query f next (use a so-called **acquisition function**);
2. Evaluate $f(\mathbf{x})$;
3. **Update the prior** based on the new data $\mathbf{x}$ and $f(\mathbf{x})$.



The prior and the acquisition function

There are several ways to specify the prior and the acquisition function. In this work, authors adopt **Gaussian Process** (GP) priors, and the **expected improvement** acquisition function.

- GP is a distribution over function specified by its mean function $m(\cdot)$ and covariance $\text{cov}(\cdot, \cdot)$. Given a set of points $\mathbf{x}_{1:t}$, with $\mathbf{x}_i \in \mathbb{R}^D$, we have

$$\mathbf{f}(\mathbf{x}_{1:t}) \sim \mathcal{N}(\mathbf{m}(\mathbf{x}_{1:t}), \mathbf{K}(\mathbf{x}_{1:t}, \mathbf{x}_{1:t})), \quad \mathbf{K}(\mathbf{x}_{1:t}, \mathbf{x}_{1:t})_{i,j} = \text{cov}(\mathbf{x}_i, \mathbf{x}_j).$$

Analytical tractability of the posterior predictive distribution at new point $\mathbf{x}$.

- Expected improvement acquisition function (exploration vs. exploitation):

$$u(\mathbf{x}|\mathcal{D}_t) = \mathbb{E} [\max\{0, f_{t+1}(\mathbf{x}) - f(\mathbf{x}^+)\} | \mathcal{D}_t],$$

where $\mathbf{x}^+ = \arg \max_{\mathbf{x} \in \{\mathbf{x}_{1:t}\}} f(\mathbf{x})$, so the next query is:

$$\mathbf{x}_{t+1} = \arg \max_{\mathbf{x} \in \mathcal{X}} u(\mathbf{x}|\mathcal{D}_t).$$

Problem

Traditional BO fails in high dimensions due to the curse of dimensionality:

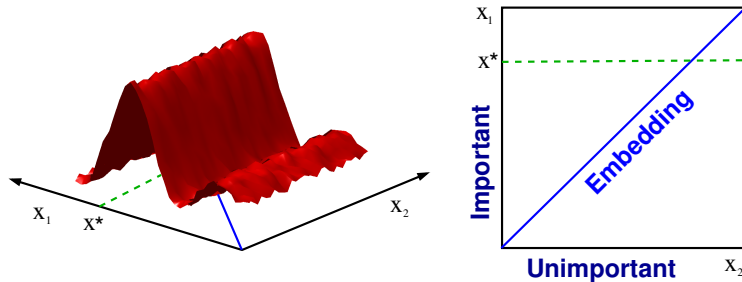
- Exponential growth in evaluations required to cover $\mathcal{X} \subset \mathbb{R}^D$.
- Most real-world problems (e.g., hyperparameter tuning, algorithm configuration) have low **effective dimensionality** $d_e \ll D$.

Key Insight

If $f(\mathbf{x})$ varies only in a d_e -dimensional subspace, we can optimize in $\mathbb{R}^d$ ($d \geq d_e$) via **random embeddings**.

Theory

Effective Dimensionality



A function $f : \mathbb{R}^D \rightarrow \mathbb{R}$ has **effective dimensionality** d_e if:

- there exists a linear subspace $\mathcal{T} \subset \mathbb{R}^D$ of dimension d_e such that $f(\mathbf{x}_{\mathcal{T}} + \mathbf{x}_{\perp}) = f(\mathbf{x}_{\mathcal{T}})$, where $\mathbf{x}_{\mathcal{T}} \in \mathcal{T}$, $\mathbf{x}_{\perp} \in \mathcal{T}^{\perp}$.
- d_e is the smallest integer with this property.

Key Theoretical Result

Theorem (Embedding Preservation)

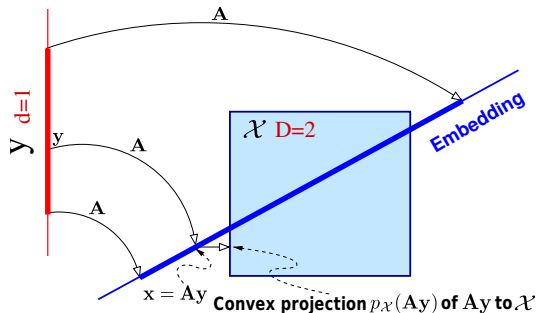
Assume we are given a function $f : \mathbb{R}^D \rightarrow \mathbb{R}$ with effective dimensionality d_e and a random matrix $\mathbf{A} \in \mathbb{R}^{D \times d}$ with independent entries sampled according to $\mathcal{N}(0, 1)$ and $d \geq d_e$. Then, with probability 1, for any $\mathbf{x} \in \mathbb{R}^D$, there exists a $\mathbf{y} \in \mathbb{R}^d$ such that $f(\mathbf{x}) = f(\mathbf{A}\mathbf{y})$.

This gives rise to Random EMbedding Bayesian Optimization (REMBO) algorithm, where we optimize the function $g(\mathbf{y}) = f(\mathbf{A}\mathbf{y})$ in the lower dimensional space.

In practice, if $\mathcal{X}$ is a compact subset (typically a box), algorithm can select a point $\mathbf{y}$ such that $\mathbf{A}\mathbf{y}$ is outside the box $\mathcal{X}$. Therefore, we can project it via $p_{\mathcal{X}} : \mathbb{R}^D \rightarrow \mathbb{R}^D$:

$$p_{\mathcal{X}}(\mathbf{y}) = \arg \min_{\mathbf{z} \in \mathcal{X}} \|\mathbf{z} - \mathbf{y}\|_2.$$

Mapping Visualization



Embedding from $d = 1$ into $D = 2$. The box illustrates the 2D constrained space $\mathcal{X}$, while the thicker red line illustrates the 1D constrained space $\mathcal{Y}$. Note that if Ay is outside $\mathcal{X}$, it is projected onto $\mathcal{X}$. The set $\mathcal{Y}$ must be chosen large enough so that the projection of its image, $A\mathcal{Y}$, onto the effective subspace (vertical axis in this diagram) covers the vertical side of the box.

Method

Bayesian Optimization vs REMBO

Algorithm 1 Bayesian Optimization

Initialize $\mathcal{D}_0$ as $\emptyset$.

for $t = 1, 2, \dots$ **do**

Find $\mathbf{x}_{t+1} \in \mathbb{R}^D$ by optimizing the acquisition function u : $\mathbf{x}_{t+1} = \arg \max_{\mathbf{x} \in \mathcal{X}} u(\mathbf{x} | \mathcal{D}_t)$.

Augment the data $\mathcal{D}_{t+1} = \mathcal{D}_t \cup \{(\mathbf{x}_{t+1}, f(\mathbf{x}_{t+1}))\}$.

Update the kernel hyper-parameters.

end

Algorithm 2 REMBO

Generate a random matrix $\mathbf{A} \in \mathbb{R}^{D \times d}$

Choose the bounded region set $\mathcal{Y} \subset \mathbb{R}^d$

Initialize $\mathcal{D}_0$ as $\emptyset$.

for $t = 1, 2, \dots$ **do**

Find $\mathbf{y}_{t+1} \in \mathbb{R}^d$ by optimizing the acquisition function u : $\mathbf{y}_{t+1} = \arg \max_{\mathbf{y} \in \mathcal{Y}} u(\mathbf{y} | \mathcal{D}_t)$.

Augment the data $\mathcal{D}_{t+1} = \mathcal{D}_t \cup \{(\mathbf{y}_{t+1}, f(\mathbf{A}\mathbf{y}_{t+1}))\}$.

Update the kernel hyper-parameters.

end

Kernel Choices

Let $K_{SE}(\mathbf{y}) = \exp(-\|\mathbf{y}\|^2/2)$. Given a length scale $\ell > 0$, we define the corresponding squared exponential kernel (**low-dimensional**) as

$$k_\ell^d(\mathbf{y}^{(1)}, \mathbf{y}^{(2)}) = K_{SE} \left(\frac{\mathbf{y}^{(1)} - \mathbf{y}^{(2)}}{\ell} \right)$$

High-Dimensional Kernel

$$k_\ell^D(\mathbf{y}^{(1)}, \mathbf{y}^{(2)}) = K_{SE} \left(\frac{p_{\mathcal{X}}(\mathbf{A}\mathbf{y}^{(1)}) - p_{\mathcal{X}}(\mathbf{A}\mathbf{y}^{(2)})}{\ell} \right)$$

Categorical Kernel (discrete spaces)

$$k_\lambda^D(\mathbf{y}^{(1)}, \mathbf{y}^{(2)}) = \exp \left(-\frac{\lambda}{2} h(s(\mathbf{A}\mathbf{y}^{(1)}), s(\mathbf{A}\mathbf{y}^{(2)}))^2 \right)$$

where h is Hamming distance, s maps to discrete space.

Hyperparameter Tuning

Balance exploration-exploitation by dynamically tuning the GP kernel length scale ℓ .

Key Mechanism

1. Initialization:

- Bounds: $\ell \in [L, U] = [0.01, 50]$.
- Threshold: $t_\sigma = 0.002$ (standard deviation threshold).

2. Exploitation Detection:

- If $\sigma(\mathbf{x}_{t+1}) < t_\sigma$ for 5 consecutive iterations, the algorithm is over-exploiting.
- Response: Decrease $U \leftarrow \alpha U$ (multiplicative decay, e.g., $\alpha = 0.8$).

3. Re-optimization:

- Re-optimize ℓ every 20 iteration (regardless of exploitation).
- Use DIRECT (local search) + CMA-ES (global search), then select the best result.

Experiments

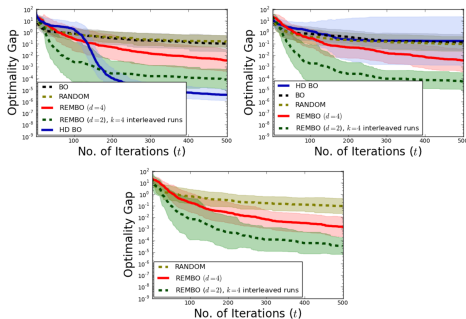
Experimental Setup

- Use function with $D = 10^9$ dimensions, but only $d_e = 2$ effective, $f(\mathbf{x}_{1:D}) = g(x_i, x_j)$, where g is the Branin function:

$$g(x_1, x_2) = \left(x_2 - \frac{5.1}{4\pi^2} x_1^2 + \frac{5}{\pi} x_1 - 6 \right)^2 + 10 \left(1 - \frac{1}{8\pi} \right) \cos(x_1) + 10.$$

- Measure the performance with the *optimality gap*.
- Fixed budget of 500 function evaluations.
- Allow to use multiple interleaved runs, for example, when using $k = 4$ interleaved REMBO runs, each of them was only allowed 125 function evaluations.
- Study the choices of k and d by considering several combinations of these values.

Comparison with other algorithms



Comparison of random search (RANDOM), Bayesian optimization (BO), method by Chen et al. (2012) (HD BO), and REMBO. Left: $D = 25$ extrinsic dimensions; Right: $D = 25$, with a rotated objective function; Bottom: $D = 109$ extrinsic dimensions. We plot means and 1/4 standard deviation confidence intervals of the optimality gap across 50 trials.

Choosing k and d

k	$d = 2$	$d = 4$	$d = 6$
10	0.0022 ± 0.0035	0.1553 ± 0.1601	0.4865 ± 0.4769
5	0.0004 ± 0.0011	0.0908 ± 0.1252	0.2586 ± 0.3702
4	0.0001 ± 0.0003	0.0654 ± 0.0877	0.3379 ± 0.3170
2	0.1514 ± 0.9154	0.0309 ± 0.0687	0.1643 ± 0.1877
1	0.7406 ± 1.8996	0.0143 ± 0.0406	0.1137 ± 0.1202

Optimality gap for $d_e = 2$ -dimensional Branin function embedded in $D = 25$ dimensions, for REMBO variants using a total of 500 function evaluations. The variants differed in the internal dimensionality d and in the number of interleaved runs k (each such run was only allowed $500/k$ function evaluations). We show mean and standard deviations of the optimality gap achieved after 500 function evaluations.

Conclusion

REMBO enables scalable Bayesian Optimization by:

1. Leveraging **random embeddings** and low intrinsic dimensionality.
2. Providing **invariance to rotations**/irrelevant dimensions.
3. Adaptively **tuning hyperparameters** for robust performance.
4. **Applications**: Hyperparameter tuning, algorithm configuration, and beyond!

Literature

1. **Main article:** Wang et al. 2013. Bayesian Optimization in a Billion Dimensions via Random Embeddings